



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Ms. Mohammed Razia Alangir Banu
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Roll Number	2520090121
Branch	CS-IT
Course Outcome	CO-2

Case Study Topic: Hotel Budget Range Query System using Segment Tree

Work Done Summary:

In this case study, a Segment Tree was implemented for managing hotel prices in the TourPlan – Smart Tourism & Travel Itinerary Optimization System. The Segment Tree efficiently processes range minimum queries to help travelers find the cheapest hotel within a specified budget range. The implementation supports fast query processing and dynamic updates, making it suitable for tourism and booking applications where hotel prices change frequently.

Implementation Details:

1. Created an array to store hotel prices.
2. Constructed a Segment Tree from the hotel price data.
3. Implemented range minimum query operations.
4. Allowed users to search hotel prices within a specified range.
5. Supported price updates without rebuilding the entire structure.
6. Compared Segment Tree performance with linear search.
7. Verified query execution efficiency.
8. Demonstrated $O(\log n)$ query and update operations.

RESULT : SCREENSHOTS/OUTPUT

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=== TOURPLAN HOTEL PRICE QUERY ===

Hotels and Prices:

Hotel Paradise -> ₹2500
Green Valley Resort -> ₹1800
Royal Stay -> ₹3200
Ocean View -> ₹1500
Hilltop Inn -> ₹2800
City Comfort -> ₹2100
Sunrise Residency -> ₹1700

Minimum Hotel Price from index 1 to 5: ₹1500

Updating Ocean View price from ₹1500 to ₹1400

New Minimum Hotel Price from index 1 to 5: ₹1400

Segment Tree Query Executed Successfully
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This case study implements a Segment Tree for efficient hotel budget range queries in the TourPlan system. The Segment Tree enables fast range minimum searches and supports dynamic updates with $O(\log n)$ complexity. The results demonstrate that Segment Trees are highly effective for tourism applications involving interval-based price analysis.

Time Complexity:

Segment Tree Construction : $O(n)$

Range Query : $O(\log n)$

Update Operation : $O(\log n)$

GitHub link: https://github.com/Jhanasri-thota/DSA_CASE-STUDIES

Student Signature	Faculty Signature